

UNSTRING

UNSTRING Syntax.

UNSTRING SourceStr\$i

[DELIMITED BY [ALL] Delim\$il [OR [ALL] Delim\$il] ...]

INTO {DestStr\$i [DELIMITER IN HoldDelim\$i]

[COUNT IN CharCounter#i]} ...

[WITH POINTER Pointer#i]

[TALLYING IN DestCounter#i]

[ON OVERFLOW StatementBlock]

[NOT ON OVERFLOW StatementBlock]

[END-UNSTRING]

```
UNSTRING FullName DELIMITED BY ALL SPACES
      INTO FirstName, SecondName, Surname
END-UNSTRING
```

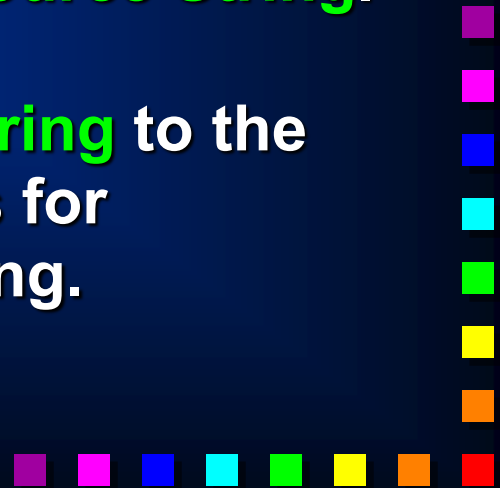
```
UNSTRING CustAddress DELIMITED BY ", "
      INTO AdrLine(1), AdrLine(2), AdrLine(3),
           AdrLine(4), AdrLine(5), AdrLine(6)
      TALLYING IN AdrLinesUsed
END-UNSTRING.
```

How the UNSTRING works

The UNSTRING copies characters from the **Source String** to the **Destination String** until a **Delimiter** is encountered in the **Source String** **or** the **Destination String** is full.

When either of these things happen the next **Destination String** becomes the receiving area and characters are copied into it until it too is full or another **Delimiter** is encountered in the **Source String**.

Characters are copied from the **Source String** to the **Destination Strings** according to the rules for Alphanumeric moves. There **is** space filling.



UNSTRING Termination

The UNSTRING statement terminates when:-

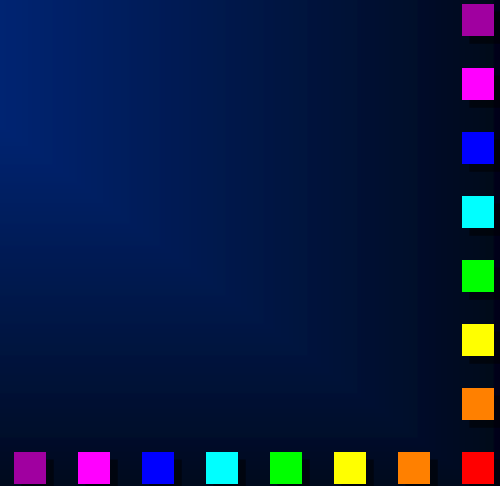
All the characters in the **Source String** have been **examined**

OR

All the **Destination Strings** have been **processed**

OR

Some **error** condition is **encountered**.



UNSTRING clauses.

◆ ON OVERFLOW.

The ON OVERFLOW is activated if :-

- ❑ The Unstring pointer (Pointer#i) is **not pointing** to a character position **within** the **SourceString** when the UNSTRING executes.
- ❑ All the Destination Strings have been processed but there are still valid **unexamined characters in the Source String**.

◆ COUNT IN

The COUNT IN clause is associated with a particular Destination String and holds a **count** of the number of **characters** passed to the Destination String.

◆ TALLYING IN

Only one TALLYING clause can be used with each UNSTRING. It holds a **count** of the number of **Destination Strings** affected by the UNSTRING operation.



The UNSTRING clauses 2.

◆ WITH POINTER

The **Pointer#i** holds the **position** of the next non-delimiter character to be **examined** in the **Source String**.

Pointer#i must be **large enough** to hold a value **one greater** than the size of the **Source String**.

◆ DELIMITER IN

A **DELIMITER IN** clause is associated with a particular Destination String. **HoldDelim\$i** holds the **Delimiter** that was **encountered** in the **Source String**.

◆ ALL

When the ALL phrase is used, **contiguous delimiters** are treated as if only **one delimiter** had been encountered.



UNSTRING Example 1

01 DayStr PIC XX.

--	--

01 MonthStr PIC XX.

--	--

01 YearStr PIC XX.

--	--

01 DateStr PIC X(8).

1	9	-	0	5	-	8	0
---	---	---	---	---	---	---	---

ACCEPT DateStr.

UNSTRING DateStr

INTO DayStr, MonthStr, YearStr

ON OVERFLOW DISPLAY "Chars Left"

END-UNSTRING.

--

UNSTRING Example 1

01 DayStr PIC XX.

1	9
---	---

01 MonthStr PIC XX.

--	--

01 YearStr PIC XX.

--	--

01 DateStr PIC X(8).

1	9	-	0	5	-	8	0
---	---	---	---	---	---	---	---

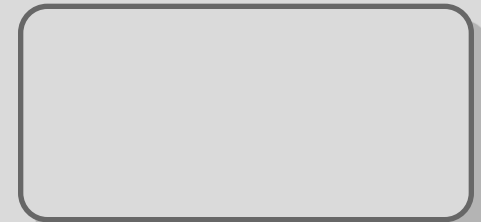
ACCEPT DateStr.

UNSTRING DateStr

INTO DayStr, MonthStr, YearStr

ON OVERFLOW DISPLAY "Chars Left"

END-UNSTRING.



UNSTRING Example 1

01 DayStr PIC XX.

1	9
---	---

01 MonthStr PIC XX.

-	0
---	---

01 YearStr PIC XX.

--	--

01 DateStr PIC X(8).

1	9	-	0	5	-	8	0
---	---	---	---	---	---	---	---

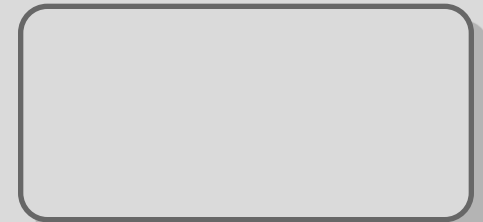
ACCEPT DateStr.

UNSTRING DateStr

INTO DayStr, MonthStr, YearStr

ON OVERFLOW DISPLAY "Chars Left"

END-UNSTRING.



UNSTRING Example 1

01 DayStr PIC XX.

1	9
---	---

01 MonthStr PIC XX.

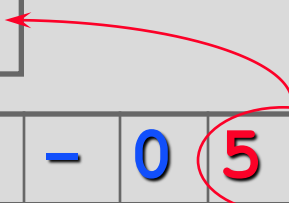
-	0
---	---

01 YearStr PIC XX.

5	-
---	---

01 DateStr PIC X(8).

1	9	-	0	5	-	8	0
---	---	---	---	---	---	---	---



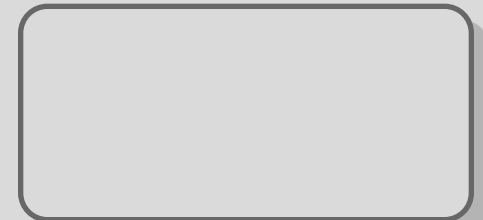
ACCEPT DateStr.

UNSTRING DateStr

INTO DayStr, MonthStr, YearStr

ON OVERFLOW DISPLAY "Chars Left"

END-UNSTRING.



UNSTRING Example 1

01 DayStr PIC XX.

1	9
---	---

01 MonthStr PIC XX.

-	0
---	---

01 YearStr PIC XX.

5	-
---	---

01 DateStr PIC X(8).

1	9	-	0	5	-	8	0
---	---	---	---	---	---	---	---

ACCEPT DateStr.

UNSTRING DateStr

INTO DayStr, MonthStr, YearStr

ON OVERFLOW DISPLAY "Chars Left"

END-UNSTRING.

Chars Left

UNSTRING Example 2

01 DayStr PIC XX.

--	--

01 MonthStr PIC XX.

--	--

01 YearStr PIC XX.

--	--

01 DateStr PIC X(8).

1	9	s	t	o	p	0	5	s	t	o	p	8	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---

ACCEPT DateStr.

UNSTRING DateStr DELIMITED BY "stop"
INTO DayStr, MonthStr, YearStr
ON OVERFLOW DISPLAY "Chars Left"
END-UNSTRING.

--

UNSTRING Example 2

01 DayStr PIC XX.

1	9
---	---

01 MonthStr PIC XX.

--	--

01 YearStr PIC XX.

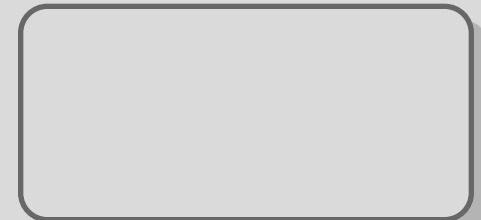
--	--

01 DateStr PIC X(8).

1	9	s	t	o	p	0	5	s	t	o	p	8	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---

ACCEPT DateStr.

UNSTRING DateStr DELIMITED BY "stop"
INTO DayStr, MonthStr, YearStr
ON OVERFLOW DISPLAY "Chars Left"
END-UNSTRING.



UNSTRING Example 2

01 DayStr PIC XX.

1	9
---	---

01 MonthStr PIC XX.

0	5
---	---

01 YearStr PIC XX.

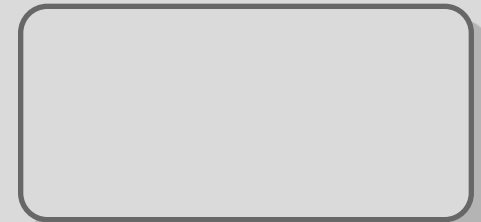
--	--

01 DateStr PIC X(8).

1	9	s	t	o	p	0	5	s	t	o	p	8	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---

ACCEPT DateStr.

UNSTRING DateStr DELIMITED BY "stop"
INTO DayStr, MonthStr, YearStr
ON OVERFLOW DISPLAY "Chars Left"
END-UNSTRING.



UNSTRING Example 2

01 DayStr PIC XX.

1	9
---	---

01 MonthStr PIC XX.

0	5
---	---

01 YearStr PIC XX.

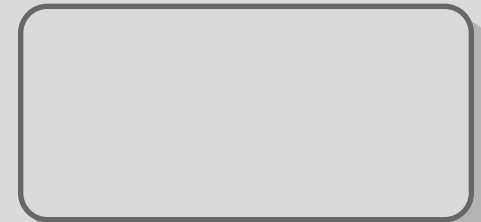
8	0
---	---

01 DateStr PIC X(8) .

1	9	s	t	o	p	0	5	s	t	o	p	8	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---

ACCEPT DateStr.

UNSTRING DateStr DELIMITED BY "stop"
INTO DayStr, MonthStr, YearStr
ON OVERFLOW DISPLAY "Chars Left"
END-UNSTRING.



UNSTRING Example 3

01 DayStr PIC XX.

1	9
---	---

01 MonthStr PIC XX.

--	--

01 YearStr PIC XX.

--	--

01 DateStr PIC X(8).

1	9	-	0	5	/	8	0
---	---	---	---	---	---	---	---

ACCEPT DateStr.

UNSTRING DateStr

DELIMITED BY "/" OR "-"

INTO DayStr DELIMITER IN Hold1

MonthStr DELIMITER IN Hold2

YearStr

END-UNSTRING.

DISPLAY DayStr SPACE MonthStr SPACE YearStr.

DISPLAY Hold1 SPACE Hold2



UNSTRING Example 3

01 DayStr PIC XX.

1	9
---	---

01 MonthStr PIC XX.

0	5
---	---

01 YearStr PIC XX.

--	--

01 DateStr PIC X(8).

1	9	-	0	5	/	8	0
---	---	---	---	---	---	---	---

ACCEPT DateStr.

UNSTRING DateStr

DELIMITED BY "/" OR "-"

INTO DayStr DELIMITER IN Hold1

MonthStr DELIMITER IN Hold2

YearStr

END-UNSTRING.

DISPLAY DayStr SPACE MonthStr SPACE YearStr.

DISPLAY Hold1 SPACE Hold2



UNSTRING Example 3

01 DayStr PIC XX.

1	9
---	---

01 MonthStr PIC XX.

0	5
---	---

01 YearStr PIC XX.

8	0
---	---

01 DateStr PIC X(8).

1	9	-	0	5	/	8	0
---	---	---	---	---	---	---	---

ACCEPT DateStr.

UNSTRING DateStr

DELIMITED BY "/" OR "-"

INTO DayStr DELIMITER IN Hold1

MonthStr DELIMITER IN Hold2

YearStr

END-UNSTRING.

DISPLAY DayStr SPACE MonthStr SPACE YearStr.

DISPLAY Hold1 SPACE Hold2



UNSTRING Example 3

01 DayStr PIC XX.

1	9
---	---

01 MonthStr PIC XX.

0	5
---	---

01 YearStr PIC XX.

8	0
---	---

01 DateStr PIC X(8).

1	9	-	0	5	/	8	0
---	---	---	---	---	---	---	---

ACCEPT DateStr.

UNSTRING DateStr

DELIMITED BY "/" OR "-"

INTO DayStr DELIMITER IN Hold1

MonthStr DELIMITER IN Hold2

YearStr

END-UNSTRING.

DISPLAY DayStr SPACE MonthStr SPACE YearStr.

DISPLAY Hold1 SPACE Hold2

19	05	80
-	/	

UNSTRING Example 4

01 DayStr PIC XX.

--	--

01 MonthStr PIC XX.

--	--

01 YearStr PIC XX.

--	--

01 DateStr PIC X(8).

1	9	-	0	5	/	8	0
---	---	---	---	---	---	---	---

ACCEPT DateStr.

UNSTRING DateStr

DELIMITED BY "/" OR "-"

INTO DayStr DELIMITER IN Hold1

MonthStr DELIMITER IN Hold1

YearStr

END-UNSTRING.

DISPLAY DayStr SPACE MonthStr SPACE YearStr.

DISPLAY Hold1

--

UNSTRING Example 4

01 DayStr PIC XX.

1	9
---	---

01 MonthStr PIC XX.

0	5
---	---

01 YearStr PIC XX.

8	0
---	---

01 DateStr PIC X(8).

1	9	-	0	5	/	8	0
---	---	---	---	---	---	---	---

ACCEPT DateStr.

UNSTRING DateStr

DELIMITED BY "/" OR "-"

INTO DayStr DELIMITER IN Hold1

MonthStr DELIMITER IN Hold1

YearStr

END-UNSTRING.

DISPLAY DayStr SPACE MonthStr SPACE YearStr.

DISPLAY Hold1

19	05	80
/		

UNSTRING Example 5

01 DayStr PIC XX.

--	--

01 MonthStr PIC XX.

--	--

01 YearStr PIC XX.

--	--

01 DateStr PIC X(11).

1	5	-	-	-	0	7	-	-	9	4
---	---	---	---	---	---	---	---	---	---	---

```
ACCEPT DateStr.  
UNSTRING DateStr  
  DELIMITED BY ALL "-"  
  INTO DayStr, MonthStr, YearStr  
  ON OVERFLOW DISPLAY "Chars Left"  
END-UNSTRING.
```

--

UNSTRING Example 5

01 DayStr PIC XX.

1	5
---	---

01 MonthStr PIC XX.

0	7
---	---

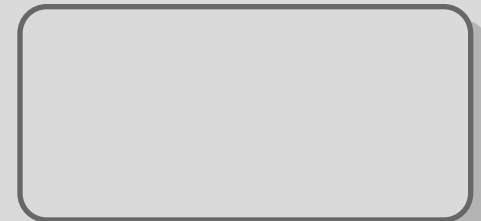
01 YearStr PIC XX.

9	4
---	---

01 DateStr PIC X(11).

1	5	-	-	-	0	7	-	-	9	4
---	---	---	---	---	---	---	---	---	---	---

```
ACCEPT DateStr.  
UNSTRING DateStr  
  DELIMITED BY ALL "-"  
  INTO DayStr, MonthStr, YearStr  
  ON OVERFLOW DISPLAY "Chars Left"  
END-UNSTRING.
```



UNSTRING Example 6

01 DayStr PIC XX.

--	--

01 MonthStr PIC XX.

--	--

01 YearStr PIC XX.

--	--

01 DateStr PIC X(11).

1	5	-	-	-	0	7	-	-	9	4
---	---	---	---	---	---	---	---	---	---	---

```
ACCEPT DateStr.  
UNSTRING DateStr  
  DELIMITED BY "-"  
  INTO DayStr, MonthStr, YearStr  
  ON OVERFLOW DISPLAY "Chars Left"  
END-UNSTRING.
```

--

UNSTRING Example 6

01 DayStr PIC XX.

01 MonthStr PIC XX.

01 YearStr PIC XX.

01 DateStr PIC X(11).

1	5
---	---

--	--

--	--

1	5	-	-	-	0	7	-	-	9	4
---	---	---	---	---	---	---	---	---	---	---

ACCEPT DateStr.

UNSTRING DateStr

DELIMITED BY "-"

INTO DayStr, MonthStr, YearStr

ON OVERFLOW DISPLAY "Chars Left"

END-UNSTRING.

Chars Left

UNSTRING Example 7

01 OldName PIC X(80).

Tim John Roberts



01 TempName.

02 NameInitial PIC X.

02 FILLER PIC X(15).

01 NewName PIC X(30).

01 Pointers.

02 StrPtr PIC 99 VALUE 1.

02 UnstrPtr PIC 99 VALUE 1.

88 NameProcessed VALUE 81.

PROCEDURE DIVISION.

ProcessName.

ACCEPT OldName.

PERFORM UNTIL NameProcessed

UNSTRING OldName DELIMITED BY ALL SPACES

INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

Display TempName

END-PERFORM

STOP RUN.

UNSTRING Example 7

```
01 OldName PIC X(80).
```

Tim John Roberts



```
01 TempName.
```

T im

```
02 NameInitial PIC X.
```

```
02 FILLER PIC X(15).
```

```
01 NewName PIC X(30).
```

```
01 Pointers.
```

```
02 StrPtr PIC 99 VALUE 1.
```

```
02 UnstrPtr PIC 99 VALUE 1.
```

```
88 NameProcessed VALUE 81.
```

Tim

```
PROCEDURE DIVISION.
```

```
ProcessName.
```

```
ACCEPT OldName.
```

```
PERFORM UNTIL NameProcessed
```

```
    UNSTRING OldName DELIMITED BY ALL SPACES
```

```
        INTO TempName WITH POINTER UnstrPtr
```

```
    END-UNSTRING
```

```
    Display TempName
```

```
END-PERFORM
```

```
STOP RUN.
```

UNSTRING Example 7

```
01 OldName PIC X(80).
```

Tim John Roberts



```
01 TempName.
```

```
02 NameInitial PIC X.
```

```
02 FILLER PIC X(15).
```

J ohn

```
01 NewName PIC X(30).
```

```
01 Pointers.
```

```
02 StrPtr PIC 99 VALUE 1.
```

```
02 UnstrPtr PIC 99 VALUE 1.
```

```
88 NameProcessed VALUE 81.
```

John

```
PROCEDURE DIVISION.
```

```
ProcessName.
```

```
ACCEPT OldName.
```

```
PERFORM UNTIL NameProcessed
```

```
UNSTRING OldName DELIMITED BY ALL SPACES
```

```
INTO TempName WITH POINTER UnstrPtr
```

```
END-UNSTRING
```

```
Display TempName
```

```
END-PERFORM
```

```
STOP RUN.
```

UNSTRING Example 7

01 OldName PIC X(80).

Tim John Roberts

01 TempName.

02 NameInitial PIC X.

02 FILLER PIC X(15).

01 NewName PIC X(30).

01 Pointers.

02 StrPtr PIC 99 VALUE 1.

02 UnstrPtr PIC 99 VALUE 1.

88 NameProcessed VALUE 81.

R oberts

Roberts

PROCEDURE DIVISION.

ProcessName.

ACCEPT OldName.

PERFORM UNTIL NameProcessed

UNSTRING OldName DELIMITED BY ALL SPACES

INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

Display TempName

END-PERFORM

STOP RUN.

UNSTRING Example 8

OldName **Tim** **John** **Roberts**



TempName

--	--

NewName

--



PROCEDURE DIVISION.

ProcessName.

ACCEPT OldName.

UNSTRING OldName DELIMITED BY ALL SPACES
 INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

PERFORM UNTIL NameProcessed

 STRING NameInitial "." DELIMITED BY SIZE
 INTO NewName WITH POINTER StrPtr

END-STRING

 UNSTRING OldName DELIMITED BY ALL SPACES
 INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

END-PERFORM

STRING SPACE TempName DELIMITED BY SIZE
 INTO NewName WITH POINTER StrPtr

END-STRING

STOP RUN.

UNSTRING Example 8

OldName **Tim**  **John** **Roberts**



TempName **T** **im**

NewName



PROCEDURE DIVISION.

ProcessName.

ACCEPT OldName.

UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr
END-UNSTRING

PERFORM UNTIL NameProcessed
STRING NameInitial "." DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr
END-STRING
UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr
END-UNSTRING

END-PERFORM

STRING SPACE TempName DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr

END-STRING

STOP RUN.

UNSTRING Example 8

OldName **Tim John Roberts**

TempName **T**im

NewName **T.**

PROCEDURE DIVISION.

ProcessName.

ACCEPT OldName.

UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

PERFORM UNTIL NameProcessed

STRING NameInitial "." DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr

END-STRING

UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

END-PERFORM

STRING SPACE TempName DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr

END-STRING

STOP RUN.

UNSTRING Example 8

OldName **Tim** **John**  **Roberts**



TempName **J** **ohn**

NewName **T** .



PROCEDURE DIVISION.

ProcessName.

ACCEPT OldName.

UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

PERFORM UNTIL NameProcessed

STRING NameInitial "." DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr

END-STRING

UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

END-PERFORM

STRING SPACE TempName DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr

END-STRING

STOP RUN.

UNSTRING Example 8

OldName **Tim** **John** **Roberts**



TempName **J** **ohn**

NewName **T.** **J.**



PROCEDURE DIVISION.

ProcessName.

ACCEPT OldName.

UNSTRING OldName DELIMITED BY ALL SPACES

INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

PERFORM UNTIL NameProcessed

STRING NameInitial "." DELIMITED BY SIZE

INTO NewName WITH POINTER StrPtr

END-STRING

UNSTRING OldName DELIMITED BY ALL SPACES

INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

END-PERFORM

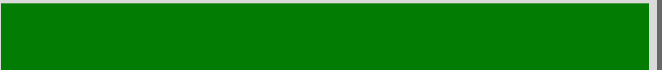
STRING SPACE TempName DELIMITED BY SIZE

INTO NewName WITH POINTER StrPtr

END-STRING

STOP RUN.

UNSTRING Example 8

OldName **Tim** **John** **Roberts** 

TempName **R** **oberts**

NewName **T.J.**

PROCEDURE DIVISION.

ProcessName.

ACCEPT OldName.

UNSTRING OldName DELIMITED BY ALL SPACES

INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

PERFORM UNTIL NameProcessed

STRING NameInitial "." DELIMITED BY SIZE

INTO NewName WITH POINTER StrPtr

END-STRING

UNSTRING OldName DELIMITED BY ALL SPACES

INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

END-PERFORM

STRING SPACE TempName DELIMITED BY SIZE

INTO NewName WITH POINTER StrPtr

END-STRING

STOP RUN.

UNSTRING Example 8

OldName **Tim John Roberts**

TempName **R**oberts

NewName **T. J.** Roberts

PROCEDURE DIVISION.

ProcessName.

ACCEPT OldName.

UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

PERFORM UNTIL NameProcessed

STRING NameInitial "." DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr

END-STRING

UNSTRING OldName DELIMITED BY ALL SPACES
INTO TempName WITH POINTER UnstrPtr

END-UNSTRING

END-PERFORM

STRING SPACE TempName DELIMITED BY SIZE
INTO NewName WITH POINTER StrPtr

END-STRING

STOP RUN.